Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: The Dark Horses
b. Team leader name: Shaik Sajid Hussain
c. Problem Statement: Future of Work & Enterprise Productivity - Rationale: the decision memory for teams

Brief about the idea

Teams make hundreds of decisions in meetings, calls and chats - and lose the WHY. Debates repeat, new hires lack context, and new decisions silently contradict old ones (e.g. Sales promises a bank an on-prem install that the CTO ruled out a month earlier).

Rationale is the decision memory for teams. Upload a meeting recording, transcript, chat export or PDF. Gemini extracts every decision with its rationale, owner, rejected alternatives and a verbatim quote into a living Decision Ledger stored in Firestore.

Its Conflict Radar reasons each new decision against the entire history and flags contradictions, supersessions and refinements in real time. 'Ask why' answers questions with clickable citations, and humans resolve conflicts while full history is preserved.

Live prototype: https://rationale-1036336415666.asia-south1.run.app

Opportunities

How is it different from existing ideas? Meeting tools summarise one meeting at a time. Rationale reasons ACROSS time: it builds a structured ledger of decisions and their rationale, and detects when a new decision contradicts, supersedes or refines an old one.

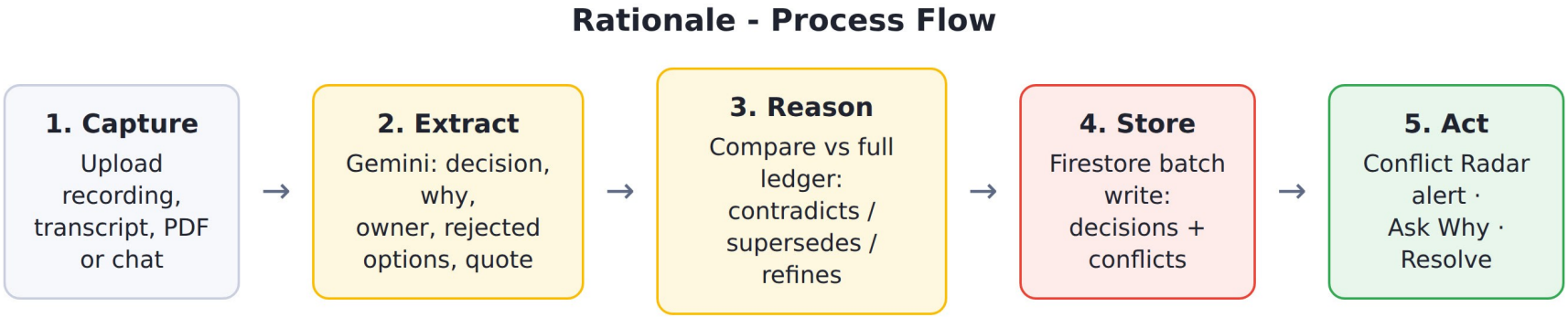
How will it solve the problem? Decisions are captured automatically from the artefacts teams already produce (recordings, transcripts, docs). Contradictions surface the moment they happen, not after a contract is signed. Anyone can ask 'why did we...?' and get a cited answer in seconds.

USP 1) Decision-level memory (what + why + who + rejected options + evidence quote). 2) Conflict Radar with conservative relation classification to avoid alert fatigue. 3) Native multimodal ingestion via Gemini (audio, PDF, text) - no separate speech-to-text. 4) Human-in-the-loop resolution with a full audit trail - valuable for regulated teams.

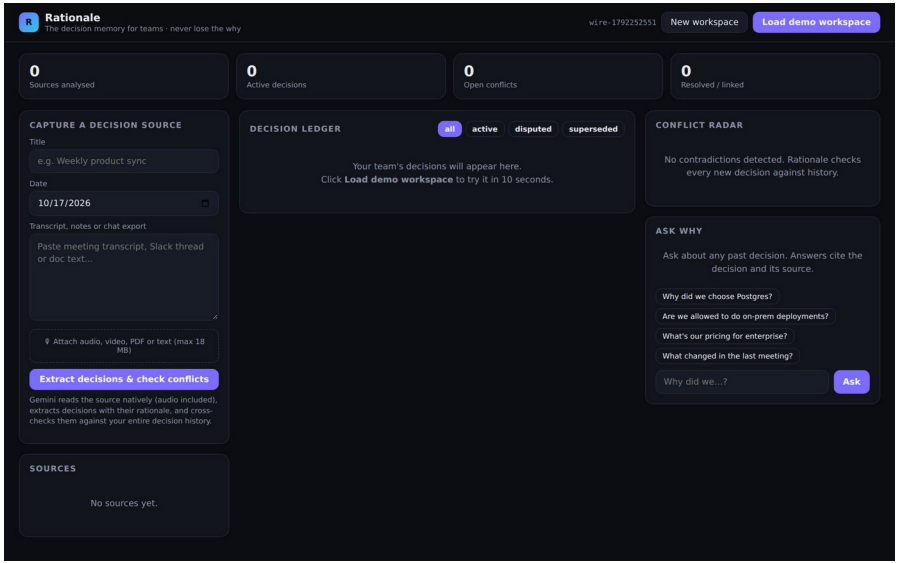
List of features offered by the solution

- Multimodal capture: paste transcripts or upload audio, video, PDF or text files (Gemini native understanding)
- Structured decision extraction: statement, rationale, owner, rejected alternatives, verbatim quote, tags
- Decision Ledger with status tracking: active / disputed / superseded, filters and D-numbers
- Conflict Radar: contradicts / supersedes / refines, with severity and plain-English explanation
- Ask Why: natural-language Q&A over the ledger with clickable [D-n] citations
- One-click conflict resolution (accept new / keep original) preserving history
- Source library with AI summaries and open questions per meeting
- Shareable workspaces (?ws=), instant demo workspace for evaluators

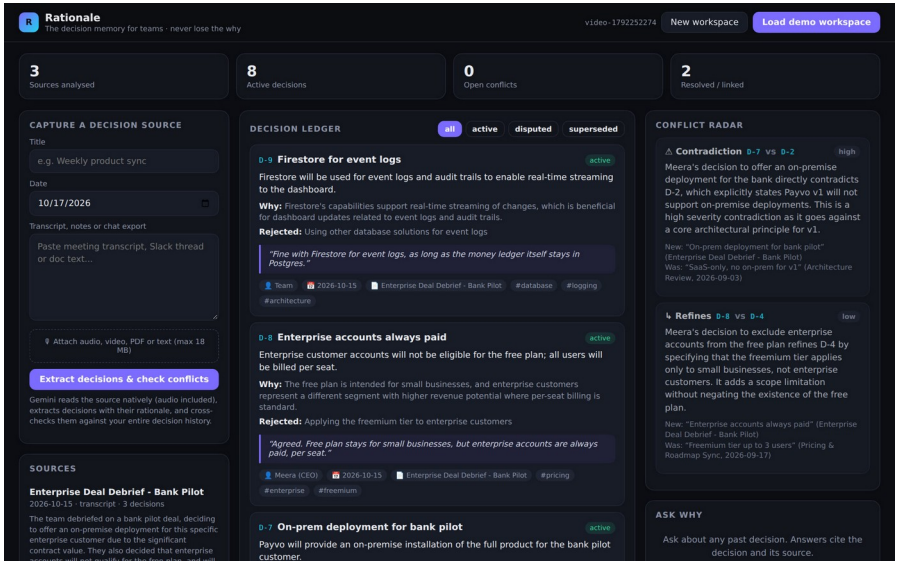
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

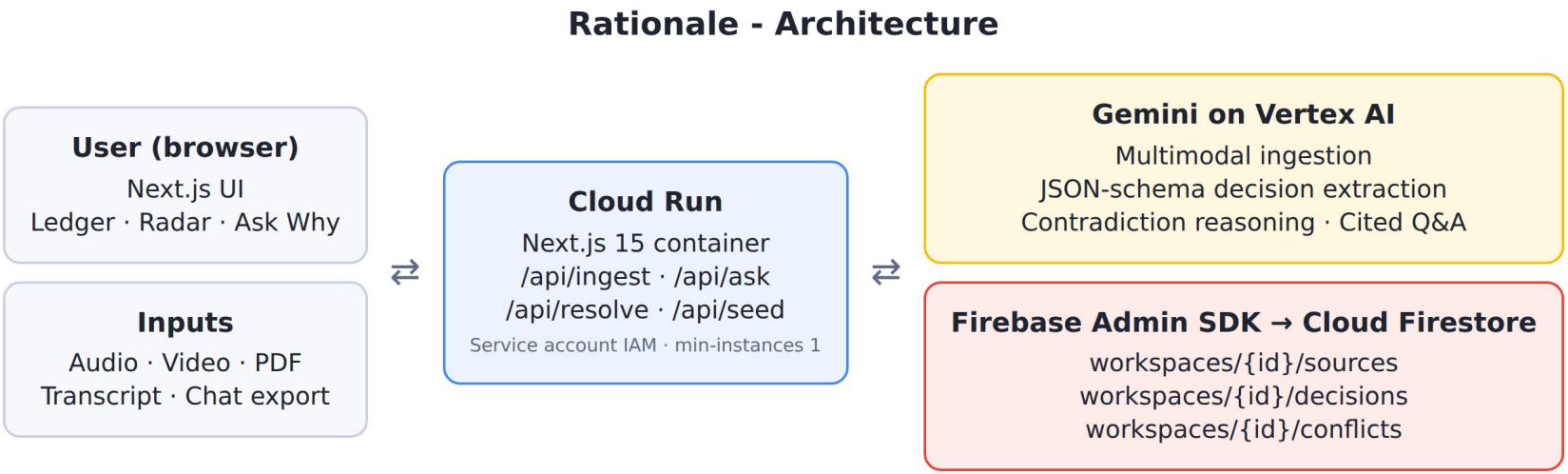


Empty workspace: capture panel, ledger, radar, ask



Populated workspace after analysis

Architecture diagram of the proposed solution



Technologies to be used in the solution

Gemini on Vertex AI - multimodal ingestion (audio/PDF/text), JSON-schema structured extraction, cross-document contradiction reasoning, cited Q&A (Google Gen AI SDK)

Firebase - Firebase Admin SDK for secure server-side access from Cloud Run using Application Default Credentials (no keys in code)

Cloud Firestore - per-workspace sources, decisions and conflicts collections; atomic batched writes keep ledger and conflict state consistent

Cloud Run - containerised Next.js 15 app (UI + API routes), autoscaling, min-instances 1 for zero cold starts, service-account IAM

Cloud Build + Artifact Registry - source-to-container builds

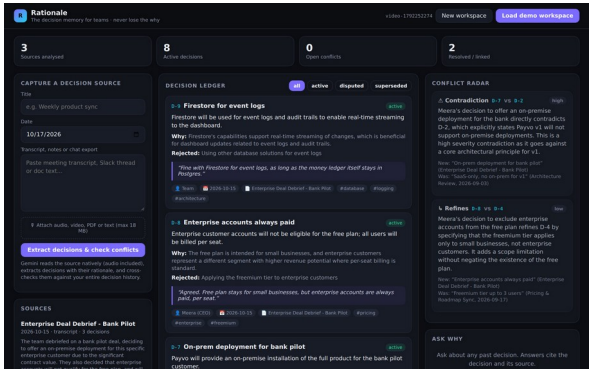
Cloud Text-to-Speech - narration for our demo video

Frontend / backend - Next.js 15, React 19, TypeScript

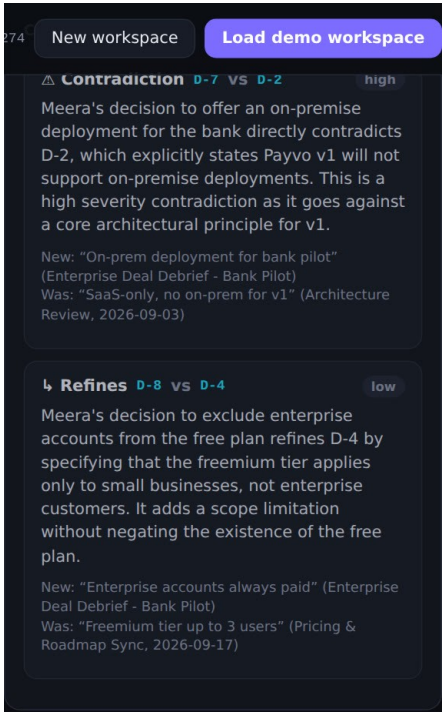
Estimated implementation cost (optional)

- Fully serverless and pay-per-use: no servers to manage, no idle VM cost.
- Cloud Run: scales with traffic; one warm instance during evaluation, can scale to zero afterwards.
- Gemini Flash-class model: one meeting analysis is a single structured call - a fraction of a cent per meeting at typical transcript sizes.
- Firestore: small documents per decision; prototype usage fits within the free tier.
- Engineering: built and deployed by a 2-person team within the hackathon window.

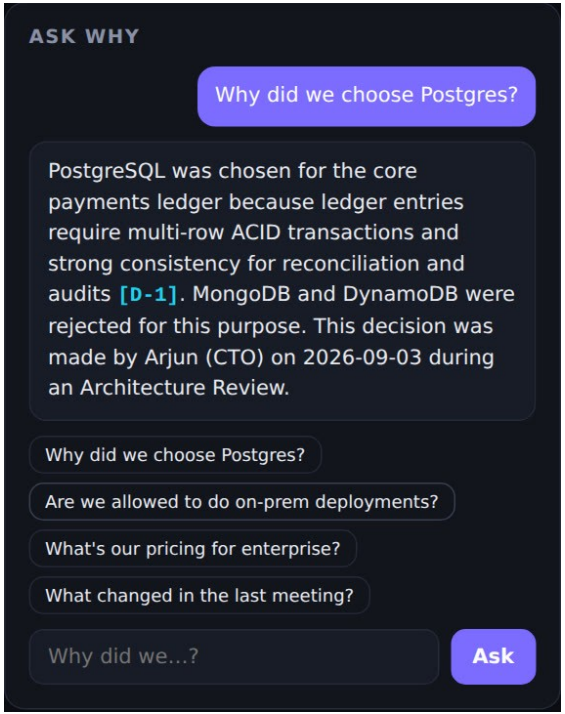
Snapshots of the prototype



Workspace overview



Conflict Radar: contradiction detected



Ask Why with citations

Prototype Performance report/Benchmarking

Measured on the live Cloud Run deployment (asia-south1) during our demo recording:

Meeting analysis (extract + reason vs full ledger, one Gemini call): 13.02 s

Ask Why answer latency: 4.1 s

Demo workspace load (Firestore batch write + read): 0.93 s

Relation detection on demo scenario: ⚠️ Contradiction D-7 vs D-2 high; ↳ Refines D-8 vs D-4 low

Precision check: unrelated decision (Firestore for event logs) correctly NOT flagged - no false positive.

Design choices for reliability: JSON-schema constrained outputs, single-call extraction+reasoning to minimise latency, atomic Firestore batches, min-instances 1 to avoid cold starts.

Additional Details/Future Development (if any)

- Connectors: Slack, Google Meet, Google Drive and Gmail to capture decisions automatically
- Firestore vector search to scale the conflict check to ledgers with 10,000+ decisions
- Weekly 'decision drift' digest for leadership
- Firebase Authentication with role-based team workspaces
- Decision templates for regulated industries (BFSI audit trails)
- Agentic follow-ups with ADK: notify decision owners automatically when their decision is contradicted

Provide links to your:

- GitHub Public Repository https://github.com/Sajidhussain19/rationale
- Demo Video Link (3 Minutes) https://youtu.be/4LVb3n2jIkE
- Final Product Link https://rationale-1036336415666.asia-south1.run.app

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Thank you

